Proposal of Adaptive Data Collection Method of On-Vehicle Sensors in the 5G Era

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***Abstract*— In the 5G era, the cellular network is designed to address various requirements from vertical industries, such as massive connectivity, high data rate and low-latency communication. Vehicle to everything (V2X) communication is considered a representative application in vertical domains. Vehicles and infrastructure can cooperate to develop a wide range of perception where sensor information from vehicles and roadside infrastructure is collected and processed at mobile edge servers. This perception information can be distributed for advanced driving support. For example, a receiver vehicle or driver extends local perception even in limited visibility. In this system, it is a key challenge to collect sensor information within the delay budget in order to provide real-time driving support. However, user experienced throughput may vary due to vehicle mobility. The quantity of sensor data that can be transmitted within the delay budget also varies. Therefore, in this paper, we propose a sensor data collection scheme with priority control and adaptive sensor set selection to maximize the quantity of sensor data under latency requirements. The study concludes that the proposed scheme is validated in a stationary environment based on 5G trials and laboratory tests.**

Keywords—5G; V2X; vehicle-infrastructure cooperation; QoS control; delay budget; Adaptive transmission; Sensor Data;

I. INTRODUCTION

In recent years, vehicles equipped with a communication unit have become widespread, and it has been possible for servers to collect and analyze traffic conditions detected by vehicle sensors on the cellular network [1-2]. In fact, a service use case has been considered in which large quantities of traffic information are collected from vehicles and sent to a server, and perception information such as a high-definition traffic map is updated and distributed [3]. The server can obtain a wide range of traffic conditions including the position of pedestrians and other vehicles approaching from the blind areas which cannot be detected by a single vehicle, and distribute the perception information to the target vehicle to support safe operation at an intersection with low visibility. In the case of such use, 5G technology [4-6] that features high data rate and low-latency communication for data collection makes it possible to distribute the perception information in real-time by collecting and analyzing a larger quantity and higher definition of sensor data. Considering the dissemination of 5G technology, we are developing an integrated analysis system which collects sensor data from vehicles and infrastructure for the traffic management and driving support as well as other purposes [7]. This system can collect and comprehensively analyze a wide range of sensor data from vehicles and roadside infrastructure and distribute the real-time traffic conditions including the behavior of pedestrians and other vehicles to them for the advanced driving support.

In the literature [7], we identified the actual end-to-end system latency with field trials using a 5G radio when the server collected the LiDAR (Laser Imaging Detection and Ranging) and camera image data from vehicles and roadside infrastructure and analyzed and distributed the perception information to provide driving support. On the other hand, user experienced throughput and latency vary due to mobility. Notwithstanding the high throughput offered by 5G, communication may be fall back to legacy 4G at least in the beginning of 5G deployment without nationwide coverage. Consequently, the quantity of sensor data that can be transmitted within the delay budget for advanced driving services also varies. The quantity of sensor data needs to be adjusted adaptively and transmitted according to throughput variation in order to fully utilize 5G capability. In addition, the transmission of more sensor data may make it possible to improve the accuracy of perception. Therefore, we conducted an initial study for the design of sensor data transmission control to track throughput variation in order to maximize the quantity of sensor data. We propose a method to define the combination of transmittable sensor data within the allowable delay considering the priority in fixed communication throughput in a static environment and we verify its validity with trials.

II. RELATED WORKS

Many experiments on vehicle-infrastructure cooperation utilizing low-latency communication have been conducted [8-9], and application of NW slicing [10] and MEC (Mobile Edge Computing) [11] to driving support systems have been studied. Also, adaptive modulation and coding (AMC) that maximizes user experienced throughput or meets the required reliability (e.g., block error rate) is commonly used in cellular communication [12]. Adaptive radio parameter control taking into account channel state information has been studied in V2X systems [13]. However, these works are for adaptive transmission control in the radio layer but transmission control in the application layer has not been incorporated. Radio aware application control under latency requirements for V2X has not been studied in depth yet. In this paper, we propose adaptive transmission control in consideration of allowable delay for achievement of a driving support system in the 5G era.

III. SYSTEM ARCHITECTURE

A. System Configuration

The integrated analysis system shown in Fig. 1 consists of the following three functions. (1) Data Collection: collection of raw sensor data from vehicles and roadside infrastructure to a server, (2) Analysis: construction of a database of perception

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information regarding moving objects detected by analysis of sensor data, (3) Distribution: distribution of the database to vehicles and notification of the approach of moving objects. The vehicle and roadside infrastructure are equipped with two types of sensors, LiDAR and camera. The server detects moving objects, like pedestrians and vehicles by analyzing the sensor data comprehensively. For certain objects, their location is detected from LiDAR data [14] and their attributes are detected from image data [15]. The server integrates the detected moving object information and static map information to construct a traffic situation map that monitors real-time traffic situations as perception information.

In a radio communication system, actual throughput variation in vehicle User Equipment (UE) is higher than that in roadside infrastructure UE due to its mobility. For this reason, the data from a vehicle in motion might not be fully transmitted depending on throughput variation in seconds, e.g., under shadowing dip. So in this paper, we focus on the data collection from the on-vehicle sensors and propose an adaptive sensor data collection method referring to the location-dependent uplink throughput map which is constructed by statistical observation of user experienced throughput at the server. The server selects several vehicles for collecting sensor data from the connected vehicles based on the traffic situation map and the uplink throughput map. In this paper, we focus on an adaptive method of collecting sensor data between one of the selected vehicles and the server. The cooperation between the server and vehicles determines selected sensor data with transmission parameters, such as, sensor set, frame rate, resolution and priority.

Fig. 2 shows the configuration of the integrated analysis system in the 5G trials. It is assumed to be a next-generation vehicle equipped with a large number of sensors, namely, one 16-beam LiDAR sensor (Velodyne VLP-16) and four high-definition cameras (Basler acA1920-40gc). Each sensor device is connected to a control board (sensor control board), and the sensor control board is connected to a 1G Ethernet switch (1G ESW). They are aggregated into 5G terminal via multi-Giga (2.5G) Ethernet switches (MG ESW). The above system configuration assumes a future vehicle network of the 5G era in which a large number of sensors physically separated at each location of the vehicle are connected by multistage connections of 1G ESWs and higher-speed MG ESWs. The sensor control board uploads the collected sensor data to the server via the 5G radio. This protocol is HTTP as an initial study. In the integrated analysis server, collected sensor data is analyzed per frame basis. Regarding the sensor data size per frame, the resolution of

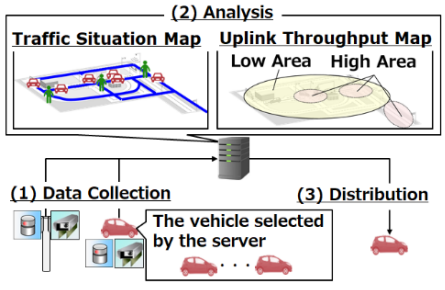


Fig.1 Outline of the integrated analysis system

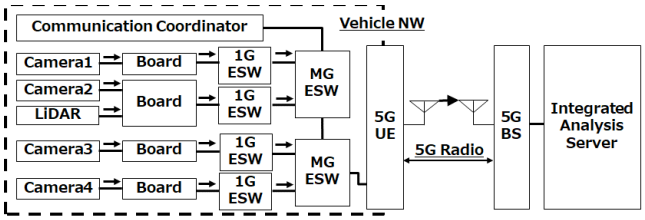


Fig.2 System configuration in 5G trials

LiDAR data is Point cloud (16×1800 pts, 91.7 KB). The resolution of the image can be selected in SD (720×480 pxs, 345.6 KB), HD (1280×720 pxs, 921.6 KB) and Full HD (1920×1080 pxs, 2073.6 KB). Since the appropriate compression rate of the image data depends on the analysis capability and processing content of the receiving server, the image data is transmitted in raw format (uncompressed) to be applicable to more general analysis applications in this paper. Transmission duration is calculated as sensor data size divided by user experienced throughput. Transmission delay was measured in the application layer from the start of transmission on the sensor control board to the completion of reception on the server as data collection time. The sensor data upload process of each sensor control board to the server is controlled by the communication coordinator. The communication coordinator grasps the throughput by using the uplink throughput map distributed from the server, and instructs a sensor dataset to be transmitted to each sensor control board according to the throughput. Each sensor control board captures sensor data and transmits it to the server with reference to the instructed dataset. The communication coordinator synchronizes each sensor control board by Network Time Protocol (NTP). The sensor dataset transmitted by each sensor control board is shown in TABLE I. We evaluated a simple case with limited number of 6 sensor dataset for an initial study. LiDAR data with small quantity of data is transmitted at a uncompressed and fixed frame rate, and the quantity of camera image data is adjusted according to the throughput. The transmission frame rate to the server is switched to 3 or 10 fps. These values are set based on the lower and the upper limit of the analysis process in this system. Regarding the correspondence between the 5G uplink data rate and sensor dataset ID in the trial, among the configurable data rates in the 5G experimental equipment, numerical values that exceed the data quantity of each sensor dataset are set. As an exception, a single uplink rate of 315 Mbps is assigned to two dataset ID = 4, 5. This assumes the case where the dataset is switched according to traffic conditions.

B. Allowable delay for data collection time

In order to define the allowable delay for sensor data collection time (DCT) for the target value in this paper, we refers to the evaluation results of [7]. In [7], a requirement for system latency (SL) of 100 ms was assumed for LiDAR data and 215 ms was assumed for image data. The distribution data size is small and can be completed in 10 ms or less regardless of the kind of sensor data. Therefore, the delivery time (DT) was uniformly set to 10 ms. The analysis time (AT) was set to the required value in the experimental environment. Because the system latency is the sum of DCT, AT and DT, an allowable

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TABLE I. Details of Sensor Dataset

ID	Sensor set	Frame rate [fps]	Data rate [Mbps]	5G UL rate [Mbps]
0	LiDAR	10	7.3	55
1	LiDAR, SD image x 4	10, 3	40.5	113
2	LiDAR, HD image x 4	10, 3	95.8	176
3	LiDAR, SD image x 4	10, 10	117.9	205
4	LiDAR, Full HD image x 4	10, 3	206.4	315
5	LiDAR, HD image x 4	10, 10	302.2	315

delay is defined as DCT = SL - AT - DT as shown in TABLE II. We describe a design method for sensor dataset where DCT is within the defined allowable delay.

TABLE II. Latency Requirements

Sensor Data	Requirement			Measurement
	SL [ms]	DT [ms]	DCT [ms]	AT [ms]
LiDAR	100	10	**63**	27
SD image	215		**131**	74
HD image			**120**	85
Full-HD image			**106**	99

IV. DISCUSSION BASED ON 5G TRIALS

We evaluated data collection time for each sensor dataset using a 5G connection in order to identify any dominating technical issue in sensor data collection that satisfies the allowable delay. The configurations of the Base Station (BS) and UE were the same in [7]. The BS was set at a distance of approximately 5.5 m from the UE in an indoor stationary environment, and in line of sight (LOS). Fig. 3 and 4 shows the CDF (cumulative distribution function) of data collection time for each sensor data in the case of sensor dataset ID = 0 and 2. Data collection time for LiDAR at ID = 2 increases at a CDF of 60% or more. This value is 65 ms larger at a CDF of 95% as compared with that of ID = 0 (when only LiDAR was transmitted), and does not satisfy the allowable delay (63 ms). This means that the data collection time of LiDAR increases because packet transmission is delayed due to the concurrent transmission of multiple sensor data. In addition, all four cameras do not satisfy the allowable delay (120 ms). The following stepwise approach can be considered in order to solve the above problems and achieve a sensor data transmission within the allowable delay.

(I) To transmit in order of high priority data within the allowable delay, and stop transmission of low priority data beyond the allowable delay.

(II) By further advancing the priority control of (I), to adjust data collection time between sensors to avoid collision.

In this paper, we discuss the specific method of (I) in terms of on-vehicle systems as an initial study. To achieve it, it is necessary to determine sensor dataset that defines the number of transmittable sensors that satisfy the allowable delay after applying QoS control to the ESW or 5G terminal, which is an aggregation point of sensor data. Therefore, in the next section, we propose a method to define the number of the transmittable sensors by estimating the maximum value of data collection time for each sensor data in the throughput and verify it with trials. An experimental system for verification of the proposed method is shown in Fig. 5. In this experimental system, an ESW where sensor data was aggregated had QoS control applied in order to verify the effect of simply applying QoS control to the uplink in terms of the on-vehicle system. As an initial study, QoS control conformed to port-based strict priority, which is one of the most basic schemes. Therefore, the control boards of Camera2 and the LiDAR were configured separately. It has been confirmed that the above control board division does not affect the verification results because the increase in sensor data transmission delay is dominant in ESW which is the data aggregation point. As for the priority, the highest priority was set to LiDAR, and the second highest priority was set to cameras. Among cameras, the one with the smaller number was set to have a higher priority.

The principal aim of this study is to determine appropriate transmission control of sensor data satisfying the allowable data collection time for the value of throughput, focusing on the decrease of the delay at the sensor data aggregation point. For this reason, it was determined to evaluate in the system with the variation factor reduced as much as possible. 5G connection was replaced with a wired LAN that emulated the 5G uplink rate by using bandwidth limitations at the output port of the ESW. Moreover, the control boards were replaced with PCs in order to make processing delay less likely to occur in sensor control units.

V. VERIFICATION OF PROPOSAL

A. Proposal

The condition for maximizing the data collection time of a certain sensor is when the data collection duration completely overlaps with other higher priority sensor data. Assuming strict priority-based QoS control, transmission of a certain sensor is started after completion of transmission of all higher priority sensor data. Therefore, the maximum value for the data

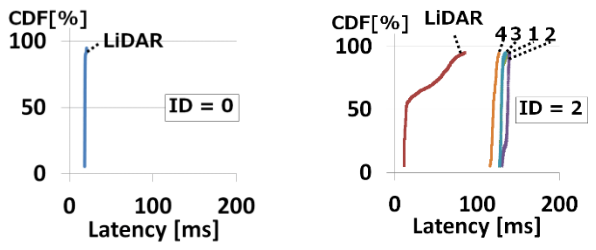


Fig. 3 and Fig. 4 Data collection time in 5G trials

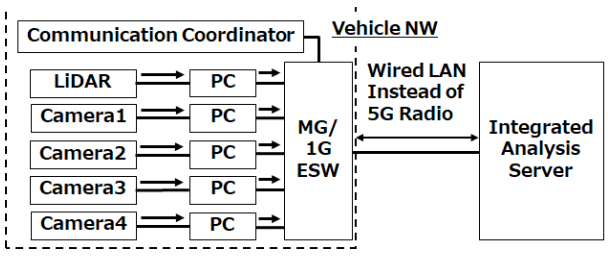


Fig.5 System composition in the trials of proposal

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collection time of a certain sensor data is formulated by the equation (1) from the sum of the maximum values of the data collection time when one kind of sensor data is transmitted alone (single transmission). Also, the number of sensors that can be transmitted within the allowable delay is defined by equation (2).

$$latency_multi(i) = \sum_{j=0}^{i} latency_sing(j) \quad (1)$$

$$i = \arg\max \{latency_multi(i) < \mathrm{L_req})\} \quad (2)$$

latency_sing(*i*) is the maximum value for the data collection time when sensor data of the *i*-th priority is transmitted alone. *latency_multi*(*i*) is the maximum value of data collection time when the sensor of the *i*-th priority is transmitted with several sensors. L_req is the value of an allowable delay. Note that *latency_sing*(*i*) is assumed to be obtained by notifying a vehicle of an observation value on a server.

B. Results and discussion

The number of transmittable cameras (Num.) calculated based on the equations (1) and (2) is added to the sensor dataset. Measured values for the maximum data collection time in the case of each sensor dataset are shown in TABLE III. Predicted values for the maximum data collection time based on equation (1) are also shown in the table. In the case of all sensor dataset, it is clear that the measured values for the maximum data collection time is within the allowable delay. Therefore, the validity of the proposed method can be confirmed. The CDF of data collection time in sensor dataset ID = 1, 2 are shown in Fig. 6-7. The CDF of low priority sensor data is stepped. The reason is that, as described above, when the sensor transmission duration overlaps, low priority sensor data transmission is queued until transmission of high priority sensor data has been done. Consequently, the data collection time of low priority sensor data increases. Moreover, since frame synchronization is not activated between sensor control units, the overlapping possibility of the data collection duration between sensors varies in each trial. Therefore, the data collection time of each dataset was verified in several trials. The study confirms that the proposed method achieves the design of transmittable sensor data within the allowable delay in a fixed throughput in a static environment.

The proposed method seems to be applicable to a vehicle driving environment by calculating statistical value, e.g., average or minimum within a fixed duration for the throughput variation during driving and selecting the dataset corresponding to that value. Further evaluation with mobility is required to validate the feasibility of the adaptive sensor collection scheme.

TABLE III. Verification results of the proposal

Rate [Mbps]	ID	Num.	Predicted [ms]	Measured [ms]	Graph
113	1	4	123.07	119.07	Fig. 6
176	2	2	101.46	96.23	Fig. 7
205	3	4	74.7	63.67	-
315	4	1	64.77	61.44	-
315	5	3	89.6	79.86	-

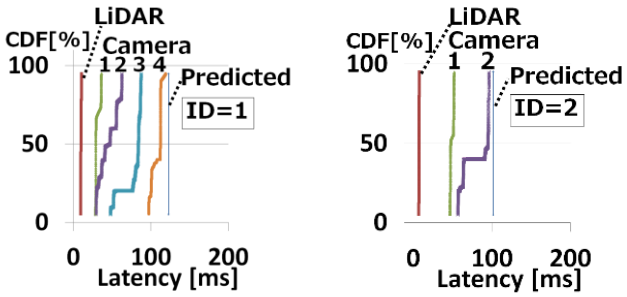


Fig. 6 and Fig. 7 Data collection time in trials of proposal

VI. CONCLUSION

In this paper, we proposed a sensor data collection scheme with priority control and adaptive sensor set selection to maximize the quantity of sensor data under latency requirements. The adaptive control is designed to operate in the 5G era. The study concluded that the proposed scheme is validated in a stationary environment based on 5G trials and laboratory tests. Future works include (1) Mobility support and (2) Scheduling of sensor data collection. Furthermore, we are studying (3) Scalability to a large number of vehicles.

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